The documentation and process conversion measures necessary to comply with this revision shall be completed by 30 June 1994

INCH-POUND

MIL-S-19500/525A
30 March 1994
SUPERSEDING
MIL-S-19500/525(USAF)
16 June 1977

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER
TYPES 2N6546, 2N6547, JAN, JANTX, AND JANTXV

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, silicon, power transistors. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See 3.3 (TO-3).

1.3 Maximum ratings.

Limits	P_T 1/		V_{CEX}	V_{CEO}	V_{EBO}	I_B	I_C	T_{OP} and T_{stg}	$R_{\theta JC}$
	T_C = +25°C	T_C = +100°C							
	W	W	V dc	V dc	V dc	A dc	A dc	°C	W/°C
2N6546	175	100	600	300	8	10	15	-65 to +200	1.0
2N6547	175	100	850	400	8	10	15	-65 to +200	1.0

1/ Between T_C = +25°C and T_C = +200°C, linear derating factor (average) = 1.0 W/°C.

1.4 Primary electrical characteristics.

Limits	h_{FE2} 1/	h_{FE3} 1/	C_{obo}	$\|h_{fe}\|$	Switching times	
	V_{CE}= 2 V dc I_C = 5 A dc	V_{CE}= 2 V dc I_C = 10 A dc	V_{CB}= 10 V dc I_E = 0 .1 MHz ≤ f ≤ 1 MHz	V_{CE}= 10 V dc I_C = 1 A dc f = 10 MHz	V_{CC}= 250 V dc I_C = 10 A dc	
					t_{on}	t_{off}
			pf			
Min	12	6		6		
Max	60		500	30	1.0	4.7

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: NASA/Parts Projects Office (NPPO), NASA Goddard Space Flight Center, Code 311.A, Greenbelt, MD 20771 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Bldg. 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, appendix F, figure 11, T-4A (TO-3).

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-S-19500.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

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4.3 Screening (JANTX and JANTXV levels). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see Table II of MIL-S-19500.	Measurement
	JANTX and JANTXV levels
10	T_A = 150°C; t = 48 hours min. V_{CB} = 200 V dc for 2N6546 V_{CB} = 300 V dc for 2N6546
11	h_{FE2}; I_{CEX1}
12	See 4.3.1
13	Subgroup 2 of table I herein. ΔI_{CEX} = 100% of initial value of 100 uA dc, whichever is greater; Δh_{FE2} = ±25% of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_J = 187.5°C ± 12.5°C; V_{CB} ≥ 20 V dc.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500. Electrical measurements (end-points) shall be in accordance with group A, subgroup 2, table I herein.

Subgroup	Method	Conditions
B3	1037	V_{CE} ≥ 20 V dc; ΔT_A between cycles = 100°C for 2,000 cycles.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end-points) shall be in accordance with group A, subgroup 2, table I herein.

Subgroup	Method	Conditions
C2	2036	Test condition A; weight = 10 pounds; time = 15 seconds.
C6	1037	V_{CE} ≥ 20 V dc; ΔT_A between cycles = 100°C for 6,000 cycles.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Collector to emitter breakdown voltage	3011	Bias condition D; I_C = 100 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N6546				300		V dc
2N6547				400		V dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 8 V dc	I_{EBO}		1.0	mA dc
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = 1.5 V dc;	I_{CEX1}		1.0	mA dc
2N6546		V_{CE} = 600 V dc				
2N6547		V_{CE} = 850 V dc				
Base emitter voltage (saturated)	3066	Test condition A; I_C = 10 A dc; I_B = 2.0 A dc; pulsed (see 4.5.1)	$V_{BE(sat)}$		1.6	V dc
Collector to emitter saturated voltage	3071	I_C = 10 A dc; I_B = 2.0 A dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$		1.5	V dc
Collector to emitter saturated voltage	3071	I_C = 15 A dc; I_B = 3.0 A dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		5.0	V dc
Forward-current transfer ratio	3076	V_{CE} = 2 V dc; I_C = 1 A dc; pulsed (see 4.5.1)	h_{FE1}	15		
Forward-current transfer ratio	3076	V_{CE} = 2 V dc; I_C = 5 A dc; pulsed (see 4.5.1)	h_{FE2}	12	60	
Forward-current transfer ratio	3076	V_{CE} = 2 V dc; I_C = 10 A dc; pulsed (see 4.5.1)	h_{FE3}	6		
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = 1.5 V dc	I_{CEX2}		30	mA dc
2N6546		V_{CE} = 600 V dc				
2N6547		V_{CE} = 850 V dc				

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued						
Low temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 5 A dc; pulsed (see 4.5.1)	h_{FE4}	4		
Subgroup 4						
Pulse response:	3251	Test condition A except test circuit and pulse requirements per figure 1 herein.				
Turn-on time		V_{CC} = 250 V dc; I_C = 10 A dc; I_{B1} = I_{B2} = 2 A dc	t_{on}		1.0	µsec
Turn-off time		V_{CC} = 250 V dc; I_C = 10 A dc; I_{B1} = I_{B2} = 2 A dc	t_{off}		4.7	µsec
Magnitude of common emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc; I_C = .5 A dc; f = 1 MHz	$\|h_{fe}\|$	6.0	30	
Output capacitance (open circuit)	3236	V_{CB} = 10 V dc; I_E = 0; 0.1 MHz ≤ f ≤ 1.0 MHz	C_{obo}		500	pF
Subgroup 5						
Safe operating area (dc operation)	3051	T_C = +25°C; t = 1 s; 1 cycle; (see fig. 2)				
Test 1 (Both device types)		V_{CE} = 11.7 V dc; I_C = 15 A dc				
Test 2 (Both device types)		V_{CE} = 20 V dc; I_C = 8.75 A dc				
Test 3 2N6546		V_{CE} = 250 V dc; I_C = 45 mA dc				
2N6547		V_{CE} = 350 V dc; I_C = 30 mA dc				

See footnote at end of table.

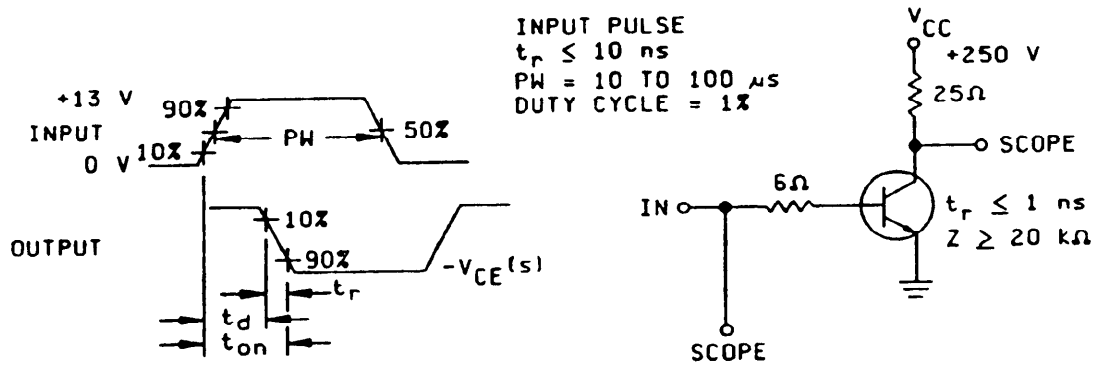
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TABLE I. Group A inspection - Continued.

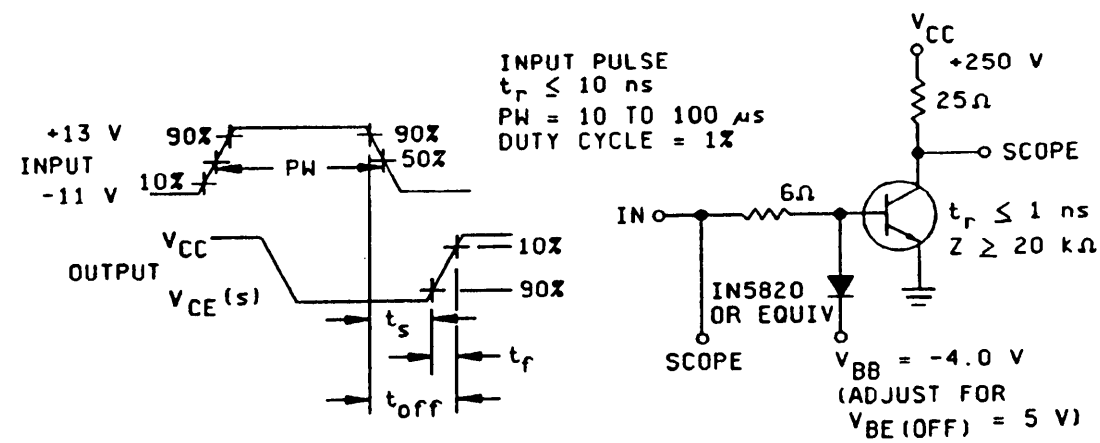
Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 5 - Continued						
Safe operating area (switching)	3053	Load condition C; (unclamped inductive load); See figure 3; T_C = 25°C; duty cycle ≤ 10%; R_S = 0.1 Ω; $t_r = t_f \leq 500$ nsec				
Test 1		t_p = 5 msec; (vary to obtain I_C); R_{BB1} = 15 Ω; V_{BB1} = 38.5 V dc; R_{BB2} = 50 Ω; V_{BB2} = -4 V dc; V_{CC} = 20 V dc; I_C = 15 A dc; L = 10 µH 2/				
Test 2		t_p = 5 msec; (vary to obtain I_C); R_{BB1} = 15 Ω; V_{BB1} = 38.5 V dc; R_{BB2} = 50 Ω; V_{BB2} = -4 V dc; V_{CC} = 20 V dc; I_C = 100 mA dc; L = 1 mH 3/				
Electrical measurements: Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = 1.5 Vdc	I_{CEX3}			
2N6546		V_{CE} = 600 Vdc		--	2.0	mA dc
2N6547		V_{CE} = 850 Vdc		--	2.0	mA dc
Safe operating area (switching)		Clamped inductive load; T_A = 25°C; duty cycle ≤ 5%; t_p = 1.5 msec; (vary to obtain I_C); V_{CC} = 20 V dc; I_C = 8 A dc; L = 180 µH (see figure 4)				
2N6546		Clamp voltage = 350 V dc				
2N6547		Clamp voltage = 450 V dc				
Electrical measurements		Same as safe operating area (unclamped inductive) above				
Subgroups 6 & 7						
Not applicable						

1/ For sampling plan, see MIL-S 19500.
2/ L = 10 µH (approx. 10 turns, 1 row of #16ASG wire on an air core 2 7/8" ID) .0007 ohms, or equivalent.
3/ L = 1 mH (1 each Miller type 7827 in parallel with 2 each series strung Miller type 7825 and this in series with 2 each series strung Miller type 7827) .45 ohms, or equivalent.

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TURN-ON (t_{on}) TIME TEST CIRCUIT



TURN-OFF (t_{off}) TIME TEST CIRCUIT

FIGURE 1. Switching time test circuits.

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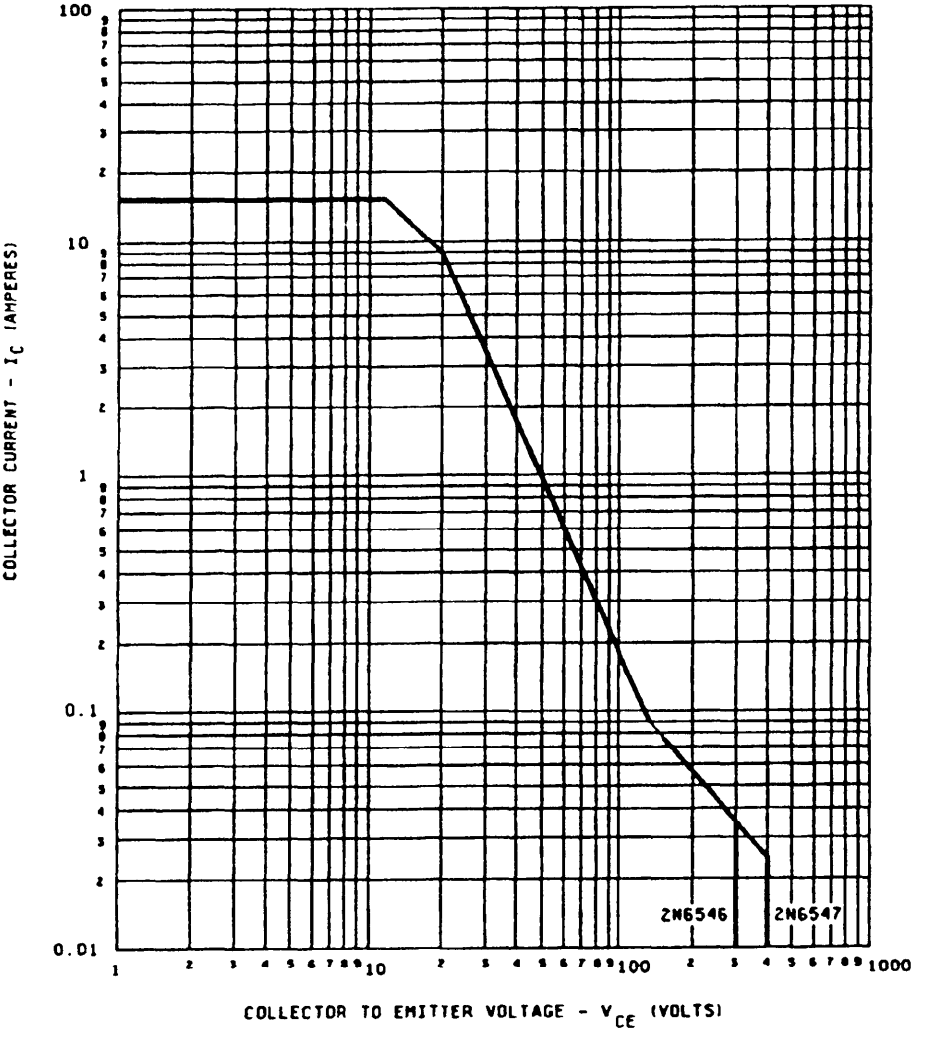


FIGURE 2. Maximum safe operating area graph (continuous DC) for types 2N6546 and 2N6547.

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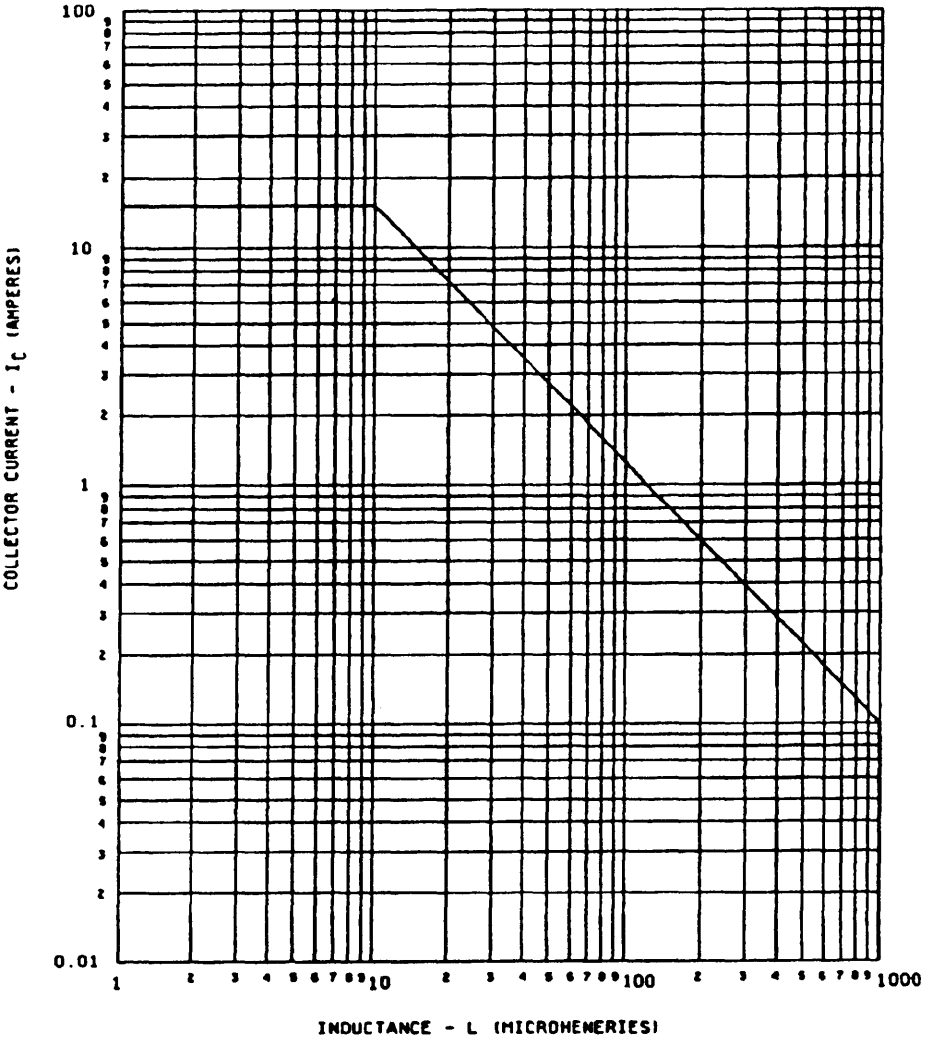
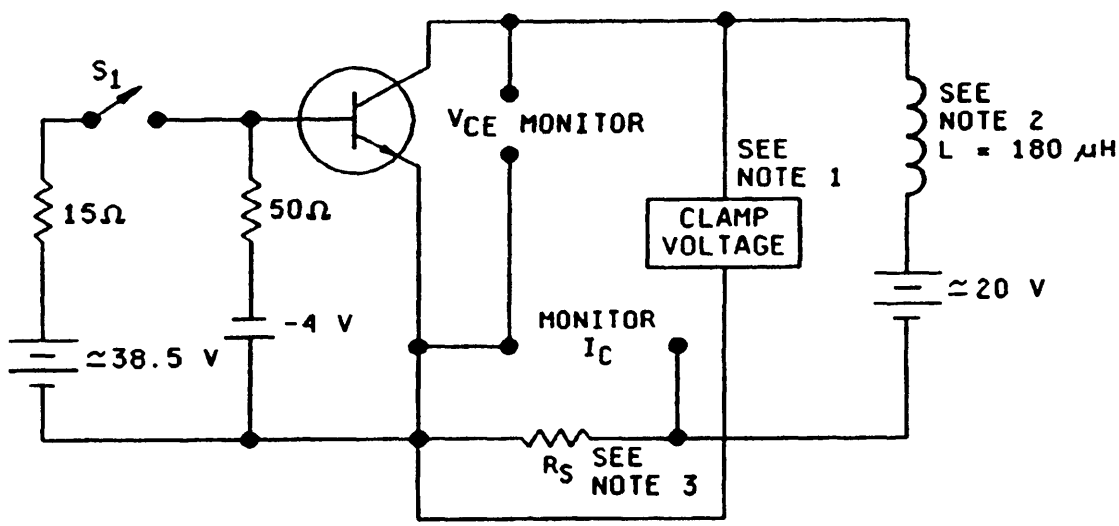


FIGURE 3. Safe operating area for switching between saturation and cutoff (unclamped inductive load).

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NOTES:
1. Either a clamping circuit or clamping diode may be used.
2. The coil used shall provide a minimum inductance of 180 μH at 8 A with a maximum dc resistance of .05 ohm. For reference only: 2 each Miller type 7827 in parallel, or equivalent.
3. $R_S \leq .1$ ohm, 12 W, 1% tolerance maximum, (noninductive).

Procedure:
1. With switch S1 closed, set the specified test conditions.
2. Open S1. Device fails if clamp voltage not reached and maintained until current reaches zero.
3. Perform specified end-point tests.

FIGURE 4. Clamped inductive sweep test circuit.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish (see 3.3.1).

c. Type designation and product assurance level.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Air Force - 17
NASA - NA

Review activities:
Air Force - 13, 19, 85, 99
DLA - ES

Preparing activity:
NASA - NA

Agent:
DLA - ES

(Project 5961-1510)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/525A	2. DOCUMENT DATE (YYMMDD) 94/03/30

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER, TYPE 2N6546, 2N6547, JANTX AND JANTXV

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (if applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial 513-296-6048	(2) AUTOVON 986-6048
c. ADDRESS (Include Zip Code) Defense Electronics Supply Center ATTN: DESC-ELDT Dayton, OH 45444-5765	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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